Chapter - 9

Reduction Formula and Beta and Gamma Functions

9.1 Reduction formula

The formula of the given integral which is connected with some integrals of lower order is called a *Reduction Formula.* It is the simplest form of the given integral. In most of the cases, the reduction formula is obtained by the method of integration by parts.

9.2 Some Important Formula

1. $$\int x^n e^{ax}\, dx = \frac{x^n e^{ax}}{a} - \frac{n}{a} I_{n-1}$$

Let $I_n = \int x^n e^{ax}\, dx$

Integrating by parts

$$I_n = x^n \frac{e^{ax}}{a} - \frac{n}{a}\int x^{n-1} e^{ax}\, dx$$

$$= x^n \frac{e^{ax}}{a} - \frac{n}{a} I_{n-1}$$

$$\therefore\ I_n = \frac{x^n e^{ax}}{a} - \frac{n}{a} I_{n-1}$$

2. $$\int \sin^n x\, dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}$$

Let $I_n = \int \sin^n x\, dx$

$$I_n = \sin^{n-1} x(-\cos x) - (n-1)\int \sin^{n-2} x \cos x\,(-\cos x)\, dx$$

$$= -\sin^{n-1} x \cos x + (n-1)\int \sin^{n-2} x \cos^2 x\, dx$$

$$= -\sin^{n-1} x \cos x + (n-1)\int \sin^{n-2} x\,(1 - \sin^2 x)\, dx$$

$$= -\sin^{n-1} x \cos x + (n-1)\int \sin^{n-2} x\, dx - (n-1)\int \sin^n x\, dx$$

$$I_n = \sin^{n-1} x \cos x + (n-1) I_{n-1} - (n-1) I_n$$

$$(1 + n - 1) I_n = -\sin^{n-1} x \cos x + (n-1) I_n$$

$$\therefore \quad I_n = -\frac{\sin^{n-1} x \cos x}{n} + \frac{(n-1)}{n} I_{n-2}$$

For example, $I_6 = \int \sin^6 x\, dx$ 2059 B.E.

By using formula

$$I_6 = -\frac{\sin^5 x \cos x}{6} + \frac{5}{6} I_4$$

$$= -\frac{\sin^5 x \cos x}{6} + \frac{5}{6}\left[-\frac{\sin^3 x \cos x}{4} + \frac{3}{4} I_2\right]$$

$$= -\frac{\sin^5 x \cos x}{6} - \frac{5}{24}\sin^3 x \cos x + \frac{5}{8} I_2$$

$$= -\frac{\sin^5 x \cos x}{6} - \frac{5}{24}\sin^3 x \cos x + \frac{5}{8}\left[-\frac{\sin x \cos x}{2} + \frac{1}{2} I_0\right]$$

But $I_0 = \int (\sin x)^0\, dx = \int dx = x$

$$\therefore \int \sin^6 x\, dx = -\frac{\sin^5 x \cos x}{6} - \frac{5}{24}\sin^3 x \cos x - \frac{5}{16}\sin x \cos x + \frac{5}{16} x$$

3. $\displaystyle \int \cos^n x\, dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} I_{n-2}$

Let $I_n = \int \cos^n x\, dx$

$$= \int \cos^{n-1} x \cos x\, dx$$

Integrating by parts

$$I_n = \cos^{n-1} x(\sin x) - (n-1)\int \cos^{n-2} x \sin x (\sin x)\, dx$$

$$= \cos^{n-1} x \sin x - (n-1)\int \cos^{n-2} x \sin^2 x\, dx$$

$$= \cos^{n-1} x \sin x - (n-1)\int \cos^{n-2} x (1 - \cos^2 x)\, dx$$

$$= \cos^{n-1} x \sin x - (n-1)\int \cos^{n-2} x\, dx + (n-1)\int \cos^n x\, dx$$

$$I_n = \cos^{n-1} x \sin x - (n-1) I_{n-1} + (n-1) I_n$$

$$(1 + n - 1) I_n = -\cos^{n-1} x \sin x + (n-1) I_n$$

$$\therefore \quad I_n = -\frac{\cos^{n-1} \sin x}{n} + \frac{(n-1)}{n} I_{n-2}$$

4. $\displaystyle\int \tan^n x\, dx = \frac{\tan^{n-1}x}{n-1} - I_{n-2}$

Let $I_n = \int \tan^n x\, dx$

$= \int \tan^{n-2} x \tan^2 x\, dx$

$= \int \tan^{n-2} x (\sec^2 x - 1)\, dx$

$I_n = \int \tan^{n-2} x \sec^2 x\, dx - \int \tan^{n-2} x\, dx$

$= \dfrac{1}{n-1} \int d(\tan^{n-1} x) - I_{n-2}$

$\therefore \quad I_n = \dfrac{\tan^{n-1} x}{n-1} - I_{n-2}$

5. $\displaystyle\int \cot^n x\, dx = -\frac{\cot^{n-1}x}{n-1} - I_{n-2}$ 2061 B.E.

Let $I_n = \int \cot^n x\, dx$

$= \int \cot^{n-2} x \cot^2 x\, dx$

$= \int \cot^{n-2} x (\operatorname{cosec}^2 x - 1)\, dx$

$I_n = \int \cot^{n-2} x \operatorname{cosec}^2 x\, dx - \int \cot^{n-2} x\, dx$

$= -\dfrac{1}{n-1} \int d(\cot^{n-1} x) - I_{n-2}$

$\therefore \quad I_n = -\dfrac{\cot^{n-1} x}{n-1} - I_{n-2}$

For example.

$I_7 = \int \cot^7 x\, dx$ 2061 B.E.

$= -\dfrac{\cot^6 x}{6} - I_5 = -\dfrac{\cot^6 x}{6} - \left[-\dfrac{\cot^4 x}{4} - I_3\right]$

$= -\dfrac{\cot^6 x}{6} + \dfrac{\cot^4 x}{4} + \left[-\dfrac{\cot^2 x}{2} - I_1\right]$

$= -\dfrac{\cot^6 x}{6} + \dfrac{\cot^4 x}{4} - \dfrac{\cot^2 x}{2} + \int -\cot x\, dx$

$$= -\frac{\cot^6 x}{6} + \frac{\cot^4 x}{4} - \frac{\cot^2 x}{2} - \log \sin x$$

6. $$\int \sec^n x\, dx = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} I_{n-2}$$

Let $I_n = \int \sec^n x\, dx$

$$= \int \sec^{n-2} x \sec^2 x\, dx$$

Integrating by parts

$$I_n = \sec^{n-2} x \tan x - \int (n-2) \sec^{n-3} x \sec x \tan x \tan x\, dx$$

$$= \sec^{n-2} x \tan x - (n-2) \int \sec^{n-2} x (\sec^2 x - 1)\, dx$$

or, $I_n = \sec^{n-2} x \tan x - (n-2) \int \sec^n x\, dx + (n-2) \int \sec^{n-2} x\, dx$

or, $I_n = \sec^{n-2} x \tan x - (n-2) I_n + (n-2) I_{n-2}$

or, $(1 + n - 2) I_n = \sec^{n-2} x \tan x + (n-2) I_{n-2}$

$\therefore \quad I_n = \dfrac{\sec^{n-2} x \tan x}{(n-1)} + \dfrac{(n-2)}{(n-1)} I_{n-2}$

For example,

Let $I_5 = \int \sec^5 x\, dx$

Applying reduction formula,

$$I_5 = \frac{1}{4} \sec^3 x \tan x + \frac{3}{4} I_3$$

$$= \frac{1}{4} \sec^3 x \tan x + \frac{3}{4}\left[\frac{\sec x \tan x}{2} + \frac{1}{2} I_1\right]$$

$$= \frac{1}{4} \sec^3 x \tan x + \frac{3}{8} \sec x \tan x + \frac{3}{8} \int \sec x\, dx$$

$$= \frac{1}{4} \sec^3 x \tan x + \frac{3}{8} \sec x \tan x + \frac{3}{8} \int \sec x\, dx$$

$$= \frac{1}{4} \sec^3 x \tan x + \frac{3}{8} \sec x \tan x + \frac{3}{8} \log(\sec x + \tan x)$$

7. $$\int \operatorname{cosec}^n x\, dx = -\frac{\operatorname{cosec}^{n-2} x \cot x}{n-1} + \frac{n-2}{n-1} I_{n-2}$$

Let $I_n = \int \operatorname{cosec}^n x\, dx$

$$= \int \text{cosec}^{n-2} x \,\text{cosec}^2 x \, dx$$

Integrating by parts

$$I_n = -\text{cosec}^{n-2} x \cot x - \int (n-2)\,\text{cosec}^{n-3} x \,\text{cosec}x\,(-\cot x)\cot x \, dx$$

$$= -\text{cosec}^{n-2} x \cot x + (n-2)\int \text{cosec}^{n-2} x\,(\text{cosec}^2 x - 1)\, dx$$

or, $I_n = -\text{cosec}^{n-2} x \cot x + (n-2)\int \text{cosec}^{n} x \, dx - (n-2)\int \text{cosec}^{n-2} x \, dx$

or, $I_n = -\text{cosec}^{n-2} x \cot x + (n-2)\, I_n - (n-2)\, I_{n-2}$

or, $(1 + n - 2)\, I_n = -\text{cosec}^{n-2} x \cot x + (n-2)\, I_{n-2}$

$$\therefore \quad I_n = -\frac{\text{cosec}^{n-2} x \cot x}{(n-1)} + \frac{(n-2)}{(n-1)} I_{n-2}$$

For example,

Let $I_5 = \int \text{cosec}^5 x \, dx$

Applying reduction formula

$$I_5 = -\frac{1}{4}\,\text{cosec}^3 x \cot x + \frac{3}{4} I_3$$

$$= -\frac{1}{4}\,\text{cosec}^3 x \cot x + \frac{3}{4}\left[\frac{-\text{cosec}\, x \cot x}{2} + \frac{1}{2} I_1\right]$$

$$= -\frac{1}{4}\,\text{cosec}^3 x \cot x - \frac{3}{8}\,\text{cosec}\, x \cot x + \frac{3}{8}\int \text{cosec}x \, dx$$

$$= -\frac{1}{4}\,\text{cosec}^3 x \cot x - \frac{3}{8}\,\text{cosec}x \cot x + \frac{3}{8}\int \text{cosec}x \, dx$$

$$= -\frac{1}{4}\,\text{cosec}^3 x \cot x - \frac{3}{8}\,\text{cosec}x \cot x + \frac{3}{8}\log(\text{cosec}x - \cot x)$$

8. $\displaystyle \int \cos^m x \cos nx \, dx = \frac{\cos^m x \sin nx}{m+n} + \frac{m}{m+n} I_{m-1, n-1}$

Let $I_{m,n} = \int \cos^m x \cos nx \, dx$

Integrating by parts

$$I_{m,n} = \cos^m x \frac{\sin nx}{n} - \int m \cos^{m-1} x(-\sin x)\frac{\sin nx \, dx}{n} dx$$

$$= \frac{\cos^m x \sin nx}{n} + \frac{m}{n}\int \cos^{m-1} x \, \sin x \sin nx \, dx$$

But, $\cos(nx - x) = \cos nx \cos x + \sin nx \sin x$

$$\cos(n-1)x - \cos nx \cos x = \sin nx \sin x$$

$$\therefore I_{m,n} = \frac{\cos^m x \sin nx}{n} + \frac{m}{n}\int \cos^{m-1} x\,[\cos(n-1)x - \cos nx \cos x]\,dx$$

$$= \frac{\cos^m x \sin nx}{n} + \frac{m}{n}\int \cos^{m-1} x \cos(n-1)x\,dx$$

$$- \frac{m}{n}\int \cos^m x \operatorname{con} nx\,dx$$

or, $$I_{m,n} = \frac{\cos^m x \sin nx}{n} + \frac{m}{n} I_{m-1,n-1} - \frac{m}{n} I_{m,n}$$

or, $$\left(1+\frac{m}{n}\right) I_{m,n} = \frac{\cos^m x \sin nx}{n} + \frac{m}{n} I_{m-1,n-1}$$

$$\therefore \quad I_{m,n} = \frac{\cos^m x \sin nx}{m+n} + \frac{m}{m+n} I_{m-1,n-1}$$

For example,

Let $I_{3,5} = \int \cos^3 x \cos 5x\,dx$

$$= \frac{\cos^3 x \sin 5x}{3+5} + \frac{3}{3+5} I_{2,4}$$

$$= \frac{1}{8}\cos^3 x \sin 5x + \frac{3}{8}\left[\frac{\cos^2 x \sin 4x}{2+4} + \frac{2}{2+4} I_{1,3}\right]$$

$$= \frac{1}{8}\cos^3 x \sin 5x + \frac{3}{48}\cos^2 x \sin 4x + \frac{1}{8} I_{1,3}$$

$$= \frac{1}{8}\cos^3 x \sin 5x + \frac{3}{48}\cos^2 x \sin 4x$$

$$+ \frac{1}{8}\left[\frac{\cos x \sin 3x}{1+3} + \frac{1}{1+3} I_{0,2}\right]$$

$$= \frac{1}{8}\cos^3 x \sin 5x + \frac{1}{16}\cos^2 x \sin x\, 4x + \frac{1}{32}\cos x \sin 3x + \frac{1}{32}\int \cos 2x\,dx$$

$$= \frac{1}{8}\cos^3 x \sin 5x + \frac{1}{16}\cos^2 x \sin 4x + \frac{1}{32}\cos x \sin 3x + \frac{1}{64}\sin 2x$$

9. $$\int \cos^m x \sin nx\,dx = -\frac{\cos^m x \cos nx}{m+n} + \frac{m}{m+n} I_{m-1,n-1}$$

Let $I_{m,n} = \int_0^{\pi/2} \cos^m x \sin nx\,dx$

Integrating by parts

$$I_{m,n} = \cos^m x \frac{(-\cos nx)}{n} - \int m \cos^{m-1} x (-\sin x) \frac{(-\cos nx)}{n} dx$$

$$= -\frac{\cos^m x \cos nx}{n} - \frac{m}{n} \int \cos^{m-1} x (\cos nx \sin x)\, dx \quad ....(1)$$

But, $\sin(nx - x) = \sin nx \cos x - \cos nx \sin x$

$\cos nx \sin x = \sin nx \cos x - \sin(nx - x)$

Substituting in (1)

$$I_{m,n} = -\frac{\cos^m x \cos nx}{n} - \frac{m}{n} \int \cos^{m-1} x [\sin nx \cos x - \sin(nx - x)]\, dx$$

$$= -\frac{\cos^m x \cos nx}{n} - \frac{m}{n} \int \cos^m x \sin nx\, dx + \frac{m}{n} \int \cos^{m-1} x \sin(n-1)x\, dx$$

$$= -\frac{\cos^m x \cos nx}{n} - \frac{m}{n} I_{m,n} + \frac{m}{n} I_{m-1,n-1}$$

or, $\left(1 + \frac{m}{n}\right) I_{m,n} = -\frac{\cos^m x \cos nx}{n} + \frac{m}{n} I_{m-1,n-1}$

or, $\frac{(n+m)}{n} I_{m,n} = -\frac{\cos^m x \cos nx}{n} + \frac{m}{n} I_{m-1,n-1}$

$$\therefore \quad I_{m,n} = -\frac{\cos^m x \cos nx}{n} + \frac{m}{m+n} I_{m-1,n-1}$$

10. $\displaystyle\int_0^{\pi/2} \sin^n x\, dx = \frac{n-1}{n} J_{n-2}$

or,

$$\int_0^{\pi/2} \cos^n x\, dx = \frac{n-1}{n} J_{n-2}$$

Let $J_n = \int_0^{\pi/2} \sin^n x\, dx$

$$= \int_0^{\pi/2} \sin^{n-1} x . \sin x\, dx$$

Integrating by parts

$$J_n = \left[\sin^{n-1} x (-\cos x\right]_0^{\pi/2} - \int_0^{\pi/2} (n-1) \sin^{n-2} x \cos x (-\cos x)\, dx$$

$$= 0 + (n-1)\int_0^{\pi/2} \sin^{n-2} x\,(1-\sin^2 x)\,dx$$

$$= (n-1)\int_0^{\pi/2} \sin^{n-2} x\,dx - (n-1)\int_0^{\pi/2} \sin^n x\,dx$$

$$J_n = (n-1)J_{n-2} - (n-1)J_n$$

$$(1+n-1)J_n = (n-1)J_{n-2}$$

$$\therefore\ J_n = \frac{n-1}{n} J_{n-2}$$

For example,

Let $J_6 = \int_0^{\pi/2} \sin^6 x\,dx$

$$= \frac{6-1}{6} J_4 = \frac{5}{6}\cdot\frac{(4-1)}{4} J_2$$

$$= \frac{5}{6}\cdot\frac{3}{4}\cdot\frac{(2-1)}{2} J_0 = \frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\int_0^{\pi/2} dx$$

$$= \frac{5}{16}\left[x\right]_0^{\pi/2} = \frac{5\pi}{32}.$$

11. $\mathbf{I_{m,n} = \int_0^{\pi/2} \cos^m x\ \cos nx\,dx = \left[\frac{m(m-1)}{m^2-n^2}\right] I_{m-2,\,n}}$

Let $I_{m,n} = \int_0^{\pi/2} \cos^m x\ \cos nx\,dx$

Integrating by parts,

$$I_{m,n} = \left[\frac{\cos^m x \sin nx}{n}\right]_0^{\pi/2} + \frac{m}{n}\int_0^{\pi/2} \cos^{m-1} x\ \sin x \sin nx\,dx$$

$$= 0 + \frac{m}{n}\int_0^{\pi/2} (\cos^{m-1} x\ \sin x)\sin nx\,dx$$

Again integrating by parts,

$$I_{m,n} = \frac{m}{n}\left[\cos^{m-1} x \sin x\left(-\frac{\cos nx}{n}\right)\right]_0^{\pi/2}$$

$$-\frac{m}{n}\int_0^{\pi/2}\left[(m-1)\cos^{m-2}x\ (-\sin x)\sin n\,x+\cos^{m-1}x\cos x\right]\left(-\frac{\cos nx}{n}\right)dx$$

$$=0-\frac{m\,(m-1)}{n^2}\int_0^{\pi/2}\left[\cos^{m-2}x\ (\sin^2 x)\cos n\,x\right]dx$$

$$+\frac{m}{n^2}\int_0^{\pi/2}\cos^m x\cos nx\,dx$$

$$=0-\frac{m\,(m-1)}{n^2}\int_0^{\pi/2}\left[\cos^{m-2}x\,(1-\cos^2 x)\cos n\,x\right]dx+\frac{m}{n^2}\,I_{m,n}$$

or, $$\left(1-\frac{m}{n^2}\right)I_{m,n}=-\frac{m\,(m-1)}{n^2}\int_0^{\pi/2}\cos^{m-2}x\cos nx\,dx$$

$$+\frac{m\,(m-1)}{n^2}\int_0^{\pi/2}\cos^m x\cos n\,x\,dx$$

or, $$\left(\frac{n^2-m}{n^2}\right)I_{m,n}=-\frac{m\,(m-1)}{n^2}\,I_{m-2,n}+\frac{m\,(m-1)}{n^2}\,I_{m,n}$$

or, $$\left[\frac{n^2-m}{n^2}-\frac{m(m-1)}{n^2}\right]I_{m,n}=-\frac{m\,(m-1)}{n^2}\,I_{m-2,n}$$

or, $$\left(\frac{n^2-m-m^2+m}{n^2}\right)I_{m,n}=-\frac{m\,(m-1)}{n^2}\,I_{m-2,n}$$

or, $$\left(\frac{n^2-m^2}{n^2}\right)I_{m,n}=-\frac{m\,(m-1)}{n^2}\,I_{m-2,n}$$

or, $$I_{m,n}=-\frac{m\,(m-1)}{n^2-m^2}\,I_{m-2,n}$$

$$\therefore\quad I_{m,n}=\frac{m\,(m-1)}{m^2-n^2}\,I_{m-2,n}$$

Exercise-13

1. Obtain a reduction formula for $\int_0^{\pi/2}\cos^n x\,dx$ and hence evaluate $\int_0^{\pi/2}\cos^{10}x\,dx$ **2060/062 B.E.**

2. Obtain a reduction formula for $\int \sec^n x\, dx$ and hence find $\int \sec^6 x\, dx$

3. If $I_n = \int_0^{\pi/4} \tan^n x\, dx$, show that $I_n + I_{n-2} = \frac{1}{n-1}$ and hence deduce the values of I_5.

4. If $I_{m,n} = \int_0^{\pi/2} \cos^m x \sin nx\, dx$, prove that

$$I_{m,n} = \frac{1}{m+n} + \frac{m}{m+n} I_{m-1,n-1}$$

5. Prove that $\int_0^{\pi/2} \cos^5 x \sin 3x\, dx = \frac{1}{3}$

6. Prove that $\int_0^{\pi/2} \cos^m x \sin mx\, dx = \frac{1}{2^{m+1}}\left(2 + \frac{2^2}{2} + \frac{2^3}{3} + \ldots + \frac{2^m}{m}\right)$

7. If m, n are positive integers, then show that

$$I_{m,n} = \int_a^b (x-a)^m (b-x)^n\, dx = \frac{n(b-a)}{m+n+1}\, I_{m,n-1}$$

8. Obtain a reduction formula for $\int \cos^m x \cos nx\, dx$ and hence show that $\int_0^{\pi/2} \cos^n x \cos nx\, dx = \frac{\pi}{2^{n+1}}$

Answers

1. $\frac{63\pi}{512}$

2. $\frac{\sec^{n-2}x \tan x}{n-1} + \frac{n-2}{n-1} I_{n-2}$, $\frac{\sec^4 x \tan x}{5} + \frac{4 \sec^2 x \tan x}{15} + \frac{8}{15} \tan x$

3. $\frac{1}{2} \log 2 - \frac{1}{4}$ 8. $\frac{\cos^m x \sin nx}{m+n} + \frac{m}{m+n} I_{m-1,n-1}$

9.3 Beta and Gamma Functions

The integrals

$$\int_0^1 x^{m-1}(1-x)^{n-1}\,dx,\ m>0,\ n>0 \text{ and}$$

$$\int_0^\infty e^{-x}x^{n-1}\,dx,\ n>0$$

are called *First Eulerian Integral* (or Beta function) and *Second Eulerian Integral* (or Gamma function) respectively and denoted by $\beta(m, n)$ and $\Gamma(n)$, we write,

$$\beta(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx,\ m>0,\ n>0$$

and $\Gamma(n) = \int_0^\infty e^{-x}x^{n-1}\,dx,\ n>0$ respectively.

9.4 Some Important Formulae

1. **$\beta(m, n) = \beta(n, m)$**

$$\beta(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$$

By using properties of definite integral

$$\beta(m,n) = \int_0^1 (1-x)^{m-1}[1-(1-x)]^{n-1}\,dx$$

or, $\beta(m,n) = \int_0^1 x^{n-1}(1-x)^{m-1}\,dx = \beta(n, m)$

$\therefore\ \beta(m, n) = \beta(n, m)$

2. **$\Gamma(1) = 1$**

$$\Gamma(1) = \int_0^\infty e^{-x}x^{1-1}\,dx = \int_0^\infty e^{-x}\,dx$$

$$= \left[-e^{-x}\right]_0^\infty = -0+1 = 1.$$

$\therefore\ \Gamma(1) = 1$

3. **$\Gamma(n+1) = n\,\Gamma(n)$ for n is any positive number.**

$$\Gamma(n+1) = \int_0^\infty e^{-x}x^{n+1-1}\,dx = \int_0^\infty e^{-x}x^{n}\,dx$$

$$= -\left[x^n e^{-x}\right]_0^\infty + n\int_0^\infty x^{n-1} e^{-x}\,dx$$

$$= 0 + n\Gamma(n) = n\Gamma(n) \qquad \left(\because \lim_{x\to\infty} \frac{x^n}{e^x} = 0\right)$$

$\therefore\ \Gamma(n+1) = n\,\Gamma(n)$

4. **$\Gamma(n+1) = n!$ for n is any positive integer.**

We have,

$\Gamma(n+1) = n\Gamma(n)$ for any positive number.

$$= n(n-1)\,\Gamma(n-1)$$

$$= n(n-1)(n-2)\,\Gamma(n-2)$$

By this process

$$\Gamma(n+1) = n(n-1)(n-2)\,........\,2.1\Gamma(1)$$

$$= n(n-1)(n-2)\,............\,2.1.1 \qquad (\because \Gamma(1) = 1)$$

$\therefore\ \Gamma(n+1) = n!$ for n is any positive integer.

5. $$\int_0^\infty e^{-kx} x^{n-1}\,dx = \frac{\Gamma(n)}{k^n}$$

Put $kx = t$, $kdx = dt$

When $x = 0$, $t = 0$, when $x = \infty$, $t = \infty$

So,

$$\int_0^\infty e^{-kx} x^{n-1}\,dx = \int_0^\infty e^{-t}\left(\frac{t}{k}\right)^{n-1} \frac{dt}{k}$$

$$= \frac{1}{k^n}\int_0^\infty e^{-t} t^{n-1}\,dt = \frac{1}{k^n}\,\Gamma(n)$$

$$\therefore\ \int_0^\infty e^{-kx} x^{n-1}\,dx = \frac{\Gamma(n)}{k^n}$$

6. $$\beta(m, n) = \frac{\Gamma(m)\,\Gamma(n)}{\Gamma(m+n)}$$

$$\beta(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$$

Integrating by parts

$$\beta(m, n) = \left[(1-x)^{n-1}\frac{x^m}{m}\right]_0^1 + (n-1)\int_0^1 (1-x)^{n-2}\frac{x^m}{m}\,dx$$

$$= 0 + \frac{n-1}{m}\int_0^1 (1-x)^{n-2}x^m\,dx$$

Again integrating by parts, we get

$$\beta(m, n) = \frac{(n-1)}{m}\frac{(n-2)}{m+1}\int_0^1 (1-x)^{n-3}x^{m+1}\,dx$$

By this process, we get

$$\beta(m, n) = \frac{(n-1)(n-2)(n-3)\,.....\,2.1}{m(m+1)(m+2)\,..\,(m+n-2)}\int_0^1 (1-x)^{n-n}x^{m+n-2}\,dx$$

$$= \frac{(n-1)!}{m(m+1)(m+2)\,..........\,(m+n-2)}\left[\frac{x^{m+n-1}}{(m+n-1)}\right]_0^1$$

$$= \frac{(n-1)!\,1.2.3\,..........\,(m-2)(m-1)}{1.2.3\,.......\,(m-2)(m-1)\,m\,(m+1)(m+2)\,......(m+n-2)(m+n-1)}$$

$$= \frac{(n-1)!\,(m-1)!}{(m+n-1)!} = \frac{\Gamma(n)\,\Gamma(m)}{\Gamma(m+n)}$$

$$\therefore \quad \beta(m, n) = \frac{\Gamma(m)\,\Gamma(n)}{\Gamma(m+n)}.$$

7. $\displaystyle \boldsymbol{\beta(m, n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\,dx = \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}}\,dx}$

We know $\displaystyle \beta(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx$

Put $x = \dfrac{1}{1+y}$, $\quad dx = -\dfrac{1}{(1+y)^2}\,dy$

When $x = 0$, $y = \infty$, when $x = 1$, $y = 0$

$$\therefore \beta(m, n) = \int_\infty^0 \left(\frac{1}{1+y}\right)^{m-1}\left(1 - \frac{1}{1+y}\right)^{n-1}\cdot\frac{-1}{(1+y)^2}\,dy$$

$$= \int_0^\infty \frac{1}{(1+y)^{m+1}}\frac{(y)^{n-1}}{(1+y)^{n-1}}\,dy = \int_0^\infty \frac{y^{n-1}}{(1+y)^{m+n}}\,dy$$

$$\therefore \quad \beta(m, n) = \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}}\,dx$$

Similarly if we write $\beta(m, n) = \beta(n, m)$,then

$$\beta(m, n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\, dx$$

$$\therefore \quad \beta(m, n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\, dx = \int_0^\infty \frac{x^{n-1}}{(1+x)^{m+n}}\, dx$$

8. $\boldsymbol{\Gamma(m)\,\Gamma(1-m) = \frac{\pi}{\sin m\pi},\ 0 < m < 1}$

$$\Gamma(m)\,\Gamma(1-m) = \frac{\Gamma(m)\,\Gamma(1-m)}{\Gamma(m+1-m)} = \beta(m, 1-m)$$

$$= \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+1-m}}\, dx$$

$$= \int_0^\infty \frac{x^{m-1}}{(1+x)}\, dx = \frac{\pi}{\sin m\pi} \quad \left(\because \int_0^\infty \frac{x^{n-1}}{(1+x)} = \frac{\pi}{\sin n\pi}\right)$$

$$\therefore \ \Gamma(m)\,\Gamma(1-m) = \frac{\pi}{\sin m\pi}$$

9. $\boldsymbol{\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}}$

We know $\dfrac{\Gamma(m)\,\Gamma(n)}{\Gamma(m+n)} = \beta(m, n)$.

Let $m = n = \dfrac{1}{2}$

So

$$\frac{\Gamma\left(\frac{1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma(1)} = \beta\left(\frac{1}{2}, \frac{1}{2}\right) = \int_0^1 x^{\frac{1}{2}-1}(1-x)^{\frac{1}{2}-1}\, dx$$

$$= \int_0^1 x^{-\frac{1}{2}}(1-x)^{-\frac{1}{2}}\, dx$$

Put $x = \sin^2\theta$. $dx = 2\sin\theta\cos\theta\, d\theta$

When $x = 0$, $\theta = 0$, when $x = 1$, $\theta = \pi/2$

$$= \int_0^{\pi/2} \frac{1}{\sin\theta\cos\theta} \cdot 2\sin\theta\cos\theta\, d\theta$$

$$= 2\int_0^{\pi/2} d\theta = 2\,[\theta]_0^{\pi/2} = 2 \cdot \frac{\pi}{2} = \pi$$

or $\Gamma\left(\frac{1}{2}\right).\Gamma\left(\frac{1}{2}\right) = \pi.$

$\therefore\ \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$

10. $$\int_0^{\pi/2} \sin^p\theta \cos^q\theta\, d\theta = \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\Gamma\left(\frac{p+q+2}{2}\right)},\ p,\ q > -1$$

$$\int_0^{\pi/2} \sin^p\theta \cos^q\theta\, d\theta = \int_0^{\pi/2} (\sin^2\theta)^{p/2}(1-\sin^2\theta)^{q/2}\, d\theta$$

Put $\sin^2\theta = x$, $2\sin\theta\cos\theta\, d\theta = dx$, $d\theta = \frac{dx}{x^{1/2}(1-x)^{1/2}}$

When $\theta = 0$, $x = 0$, when $\theta = \pi/2$, $x = 1$

So,

$$\int_0^{\pi/2} \sin^p\theta \cos^q\theta\, d\theta = \int_0^1 x^{\frac{p}{2}}(1-x)^{\frac{q}{2}} \frac{dx}{2x^{1/2}(1-x)^{1/2}}$$

$$= \frac{1}{2}\int_0^1 x^{\frac{p}{2}-\frac{1}{2}}(1-x)^{\frac{q}{2}-\frac{1}{2}}\, dx$$

$$= \frac{1}{2}\int_0^1 x^{\frac{p+1}{2}-1}(1-x)^{\frac{q+1}{2}-1}\, dx$$

$$= \frac{1}{2}\beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$$

$$= \frac{1}{2}\,\frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{p+q+2}{2}\right)}$$

$$\therefore\ \int_0^{\pi/2} \sin^p\theta \cos^q\theta\, d\theta = \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\Gamma\left(\frac{p+q+2}{2}\right)}$$

11. $$\int_0^{\pi/2} \sin^p\theta\, d\theta = \frac{\sqrt{\pi}}{2}\,\frac{\Gamma\left(\frac{p+1}{2}\right)}{\Gamma\left(\frac{p+2}{2}\right)}$$

If we put $q = 0$ in the above result, then we get

$$\int_0^{\pi/2} \sin^p d\theta = \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{2\Gamma\left(\frac{p+2}{2}\right)} = \frac{\sqrt{\pi}\,\Gamma\left(\frac{p+1}{2}\right)}{2\Gamma\left(\frac{p+2}{2}\right)}$$

$$\therefore \int_0^{\pi/2} \sin^p \theta\, d\theta = \frac{\sqrt{\pi}}{2}\frac{\Gamma\left(\frac{p+1}{2}\right)}{\Gamma\left(\frac{p+2}{2}\right)}$$

12. $$\int_0^{\pi/2} \cos^q \theta\, d\theta = \frac{\sqrt{\pi}}{2}\frac{\Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{q+2}{2}\right)}$$

For,

If we put p = 0 in the above result then we get,

$$\int_0^{\pi/2} \sin^q d\theta = \frac{\Gamma\left(\frac{q+1}{2}\right)\Gamma\left(\frac{1}{2}\right)}{2\Gamma\left(\frac{q+2}{2}\right)} = \frac{\sqrt{\pi}\,\Gamma\left(\frac{q+1}{2}\right)}{2\Gamma\left(\frac{q+2}{2}\right)}$$

$$\therefore \int_0^{\pi/2} \sin^q \theta\, d\theta = \frac{\sqrt{\pi}}{2}\frac{\Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{q+2}{2}\right)}$$

13. $$\int_0^{\infty} e^{-x^2} dx = \frac{1}{2}\sqrt{\pi}$$

For,

Here, $\int_0^{\infty} e^{-x^2} dx$

Put $x^2 = t$, $2xdx = dt$, $dx = \frac{dt}{2t^{1/2}}$

When $x = 0$, $t = 0$, when $x = \infty$, $t = \infty$

So, $\int_0^{\infty} e^{-x^2} dx = \int_0^{\infty} e^{-t} \frac{dt}{2t^{1/2}}$

$$= \frac{1}{2}\int_0^{\infty} e^{-t} t^{-1/2} dt = \frac{1}{2}\int_0^{\infty} e^{-t} t^{\frac{1}{2}-1} dt$$

$$= \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{1}{2}\sqrt{\pi}.$$

$$\therefore \int_0^\infty e^{-x^2}\,dx = \frac{1}{2}\sqrt{\pi}$$

Worked Out Examples

Ex. 1: Show that i. $\Gamma\left(\frac{9}{2}\right) = \frac{105}{16}\sqrt{\pi}$ ii. $\frac{\Gamma\left(\frac{7}{2}\right)\Gamma\left(\frac{3}{2}\right)}{\Gamma(6)} = \frac{\pi}{128}$

i. $\Gamma\left(\frac{9}{2}\right) = \frac{105}{16}\sqrt{\pi}$

Solution:

$$\text{Here, } \Gamma\left(\frac{9}{2}\right) = \frac{9}{2}\times\frac{5}{2}\times\frac{3}{2}\times\frac{1}{2}\times\Gamma\left(\frac{1}{2}\right)$$

$$= \frac{105}{16}\sqrt{\pi}.$$

$$\therefore \Gamma\left(\frac{9}{2}\right) = \frac{105}{16}\sqrt{\pi}$$

ii. $\frac{\Gamma\left(\frac{7}{2}\right)\Gamma\left(\frac{3}{2}\right)}{\Gamma(6)} = \frac{\pi}{128}$

Solution:

$$\text{Here } \frac{\Gamma\left(\frac{7}{2}\right)\Gamma\left(\frac{3}{2}\right)}{\Gamma(6)} = \frac{\frac{5}{2}\times\frac{3}{2}\times\frac{1}{2}\times\Gamma\left(\frac{1}{2}\right)\frac{1}{2}\Gamma\left(\frac{1}{2}\right)}{\Gamma(5+1)}$$

$$= \frac{\frac{5}{2}\times\frac{3}{2}\times\frac{1}{2}\times\sqrt{\pi}\times\frac{1}{2}\times\sqrt{\pi}}{5\times4\times3\times2\times1} = \frac{\pi}{128}$$

$$\therefore \frac{\Gamma\left(\frac{7}{2}\right)\Gamma\left(\frac{3}{2}\right)}{\Gamma 6} = \frac{\pi}{128}$$

Ex. 2: Show that $\int_0^{\pi/2} \sin^3 x \cos^5 x\,dx = \frac{1}{24}$

Solution:

$$\text{Here, } \int_0^{\pi/2} \sin^3 x \cos^5 x\,dx = \frac{\Gamma\left(\frac{3+1}{2}\right)\Gamma\left(\frac{5+1}{2}\right)}{2\Gamma\left(\frac{3+5+2}{2}\right)} = \frac{\Gamma(2)\,\Gamma(3)}{2\Gamma(5)}$$

$$= \frac{1\times2\times1}{2\times4\times3\times2\times1} = \frac{1}{24}$$

$$\therefore \int_0^{\pi/2} \sin^3x \cos^5x \, dx = \frac{1}{24}.$$

Ex. 3: Use Gamma function to evaluate $\int_0^1 x^6\sqrt{1-x^2}\,dx$

2068 Chaitra B. E.

Solution:

Given integral is

$$\int_0^1 x^6\sqrt{1-x^2}\,dx$$

Put $x^2 = t$, $2xdx = dt$, $dx = \frac{dt}{2\sqrt{t}}$

When, $x = 0$, $t = 0$, when $x = 1$, $t = 1$

$$\therefore \int_0^1 x^6\sqrt{1-x^2}\,dx = \int_0^1 \frac{t^3 (1-t)^{1/2}\,dt}{2\, t^{1/2}} = \frac{1}{2}\int_0^1 t^{5/2}(1-t)^{1/2}\,dt$$

$$= \frac{1}{2}\int_0^1 t^{\frac{7}{2}-1}(1-t)^{\frac{3}{2}-1}\,dt = \frac{1}{2}\beta\left(\frac{7}{2}, \frac{3}{2}\right)$$

$$= \frac{1}{2}\frac{\Gamma\left(\frac{7}{2}\right)\Gamma\left(\frac{3}{2}\right)}{\Gamma\left(\frac{7}{2}+\frac{3}{2}\right)} = \frac{1}{2}\frac{\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\Gamma\left(\frac{1}{2}\right)\cdot\frac{1}{2}\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{10}{2}\right)}$$

$$= \frac{\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi}\times\frac{1}{2}\sqrt{\pi}}{2\times4\times3\times2\times1} = \frac{5}{256}\sqrt{\pi}.\sqrt{\pi} = \frac{5\pi}{256}$$

$$\therefore \int_0^1 x^6\sqrt{1-x^2}\,dx = \frac{5\pi}{256}.$$

Ex. 4: Use Beta and Gamma function to evaluate $\int_0^{2a} x^5\sqrt{2ax - x^2}\, dx$

2069 /072 B. E.

Solution:

Given integral is $\int_0^{2a} x^5\sqrt{2ax - x^2}\, dx$

Put $x = 2a\sin^2\theta$, $\quad dx = 4a\sin\theta\cos\theta\, d\theta$.

When, $x = 0$, $\theta = 0$, when $x = 2a$, $\theta = \frac{\pi}{2}$

$$\therefore \int_0^{2a} x^5\sqrt{2ax - x^2}\, dx$$

$$= \int_0^{\pi/2} (2a)^5\sin^{10}\theta\sqrt{4a^2\sin^2\theta - 4a^2\sin^4\theta}\; 4a\sin\theta\cos\theta\, d\theta$$

$$= 256a^7\int_0^{\pi/2} \sin^{12}\theta\cos^2\theta\, d\theta$$

$$= 256a^7\frac{\Gamma\left(\frac{12+1}{2}\right)\Gamma\left(\frac{2+1}{2}\right)}{2\Gamma\left(\frac{12+2+2}{2}\right)}$$

$$= 256a^7\frac{\frac{11}{2}\cdot\frac{9}{2}\cdot\frac{7}{2}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi}.\frac{1}{2}\sqrt{\pi}}{2\times 7\times 6\times 5\times 4\times 3\times 2\times 1} = \frac{33\pi a^7}{16}.$$

$$\therefore \int_0^{2a} x^5\sqrt{2ax - x^2}\, dx = \frac{33\pi a^7}{16}.$$

Ex. 5: Use Gamma function to prove $\int_0^{\pi/8} \cos^3 4x\, dx = \frac{1}{6}$

Solution:

Given integral is $\int_0^{\pi/8} \cos^3 4x\, dx$

Put, $4x = \theta$, $4\,dx = d\theta$.

When $x = 0$, $\theta = 0$, when $x = \pi/8$, $\theta = \pi/2$

$$\therefore \quad \int_0^{\pi/4} \cos^3 4x\,dx = \frac{1}{4}\int_0^{\pi/4} \cos^3\theta\,d\theta = \frac{1}{4}\,\frac{\sqrt{\pi}\,\Gamma\left(\frac{3+1}{2}\right)}{2\Gamma\left(\frac{3+2}{2}\right)}$$

$$= \frac{\sqrt{\pi}}{4}\,\frac{\Gamma(2)}{2\Gamma\left(\frac{5}{2}\right)} = \frac{\sqrt{\pi}}{4}\,\frac{1}{2\times\frac{3}{2}\times\frac{1}{2}\sqrt{\pi}} = \frac{1}{6}$$

$$\therefore \int_0^{\pi/4} \cos^3 4x\,dx = \frac{1}{6}.$$

Ex. 6: Prove that $\frac{2^n}{\sqrt{\pi}}\,\Gamma\left(\frac{2n+1}{2}\right) = 1.3.5 \,.......(2n-3)(2n-1)$

Solution:

Given gamma function is

$$\frac{2^n}{\sqrt{\pi}}\,\Gamma\left(\frac{2n+1}{2}\right) = \frac{2^n}{\sqrt{\pi}}\;\Gamma\left(n+\frac{1}{2}\right)$$

$$= \frac{2^n}{\sqrt{\pi}}\left(n+\frac{1}{2}-1\right)\,\Gamma\left(n+\frac{1}{2}-1\right)$$

$$= \frac{2^n}{\sqrt{\pi}}\left(n-\frac{1}{2}\right)\,\Gamma\left(n-\frac{1}{2}\right)$$

$$= \frac{2^n}{\sqrt{\pi}}\left(n-\frac{1}{2}\right)\left(n-\frac{3}{2}\right)\Gamma\left(n-\frac{3}{2}\right)$$

$$= \frac{2^n}{\sqrt{\pi}}\left(n-\frac{1}{2}\right)\left(n-\frac{3}{2}\right)\left(n-\frac{5}{2}\right)\Gamma\left(n-\frac{5}{2}\right)$$

$$= \frac{2^n}{\sqrt{\pi}}\left(n-\frac{1}{2}\right)\left(n-\frac{3}{2}\right)\left(n-\frac{5}{2}\right)\,......\,\frac{1}{2}\,\Gamma\left(\frac{1}{2}\right)$$

$$= \frac{2^n}{\sqrt{\pi}}\,\frac{(2n-1)}{2}\,\frac{(2n-3)}{2}\,\frac{(2n-5)}{2}\,..........\,\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\,\Gamma\left(\frac{1}{2}\right)$$

$$= \frac{2^n}{\sqrt{\pi}}\,\frac{1.3.5\,...........(2n-3)(2n-1)}{2^n}\sqrt{\pi}$$

$$= 1.3.5\,.............\,(2n-3)(2n-1)$$

$\therefore \quad \frac{2^n}{\sqrt{\pi}} \Gamma\left(\frac{2n+1}{2}\right) = 1.3.5 \ldots\ldots\ldots\ldots (2n-3)(2n-1)$

Ex. 7: Prove that $\Gamma\left(\frac{1}{9}\right)\Gamma\left(\frac{2}{9}\right) \ldots\ldots\ldots \Gamma\left(\frac{8}{9}\right) = \frac{16}{3}\pi^4$

Solution:

Here, $\Gamma\left(\frac{1}{9}\right)\Gamma\left(\frac{2}{9}\right) \ldots\ldots\ldots \Gamma\left(\frac{8}{9}\right)$

$$= \Gamma\left(\frac{1}{9}\right).\Gamma\left(\frac{2}{9}\right).\Gamma\left(\frac{3}{9}\right).\Gamma\left(\frac{4}{9}\right).\Gamma\left(\frac{5}{9}\right).\Gamma\left(\frac{6}{9}\right).\Gamma\left(\frac{7}{9}\right).\Gamma\left(\frac{8}{9}\right)$$

$$= \Gamma\left(\frac{1}{9}\right)\Gamma\left(\frac{8}{9}\right).\Gamma\left(\frac{2}{9}\right)\Gamma\left(\frac{7}{9}\right).\Gamma\left(\frac{3}{9}\right)\Gamma\left(\frac{6}{9}\right).\Gamma\left(\frac{4}{9}\right)\Gamma\left(\frac{5}{9}\right)$$

$$= \Gamma\left(\frac{1}{9}\right)\Gamma\left(1-\frac{1}{9}\right).\Gamma\left(\frac{2}{9}\right)\Gamma\left(1-\frac{2}{9}\right).\Gamma\left(\frac{3}{9}\right)\Gamma\left(1-\frac{3}{9}\right)\Gamma\left(\frac{4}{9}\right)\Gamma\left(1-\frac{4}{9}\right)$$

$$= \frac{\pi}{\sin\frac{\pi}{9}}.\frac{\pi}{\sin\frac{2\pi}{9}}.\frac{\pi}{\sin\frac{3\pi}{9}}.\frac{\pi}{\sin\frac{4\pi}{9}}$$

$$= \frac{\pi^4}{\sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ}$$

$$= \frac{4\pi^4}{\sqrt{3}\sin 20^\circ\ (2\sin 40^\circ \sin 80^\circ)}$$

$$= \frac{4\pi^4}{\sqrt{3}\sin 20^\circ\ [\cos(40^\circ - 80^\circ) - \cos(40^\circ + 80^\circ)]}$$

$$= \frac{1}{\sqrt{3}\sin 20^\circ}\ \frac{4\pi^4}{(\cos 40^\circ - \cos 120^\circ)}$$

$$= \frac{1}{\sqrt{3}\sin 20^\circ}\ \frac{4\pi^4}{\left(\cos 40^\circ + \frac{1}{2}\right)} = \frac{4\pi^4}{\left(\sqrt{3}\sin 20^\circ \cos 40^\circ + \frac{\sqrt{3}}{2}\sin 20^\circ\right)}$$

$$= \frac{8\pi^4}{\sqrt{3}\,(2\sin 20^\circ \cos 40^\circ) + \sqrt{3}\sin 20^\circ}$$

$$= \frac{8\pi^4}{\sqrt{3}\,[\sin(20^\circ + 40^\circ) + \sin(20^\circ - 40^\circ)] + \sqrt{3}\sin 20^\circ}$$

$$= \frac{8\pi^4}{\sqrt{3}\sin 60^\circ - \sqrt{3}\sin 20^\circ + \sqrt{3}\sin 20^\circ} = \frac{8\pi^4}{\sqrt{3}.\frac{\sqrt{3}}{2}} = \frac{16}{3}\pi^4$$

$\therefore \quad \Gamma\left(\frac{1}{9}\right)\Gamma\left(\frac{2}{9}\right) \ldots\ldots\ldots \Gamma\left(\frac{8}{9}\right) = \frac{16}{3}\pi^4$

Ex. 8: Use Gamma Function to prove that $\int_0^1 \frac{dx}{(1-x^6)^{1/6}} = \frac{\pi}{3}$

2060/068/071Chaitra B .E.

Solution:

Given integral is $\int_0^1 \frac{dx}{(1-x^6)^{1/6}}$

Put $x^6 = t$, $6x^5\,dx = dt$, $dx = \frac{dt}{6t^{5/6}}$

When $x = 0$, $t = 0$, when, $x = 1$, $t = 1$

So, $\int_0^1 \frac{dx}{(1-x^6)^{1/6}} = \frac{1}{6}\int_0^1 \frac{dt}{(1-t)^{1/6}\,t^{5/6}} = \frac{1}{6}\int_0^1 t^{-5/6}(1-t)^{-1/6}\,dt$

$$= \frac{1}{6}\int_0^1 t^{\frac{1}{6}-1}(1-t)^{\frac{5}{6}-1}\,dt = \frac{1}{6}\,\beta\left(\frac{1}{6}, \frac{5}{6}\right)$$

$$= \frac{1}{6}\,\frac{\Gamma\left(\frac{1}{6}\right)\Gamma\left(\frac{5}{6}\right)}{\Gamma\left(\frac{1}{6}+\frac{5}{6}\right)} = \frac{1}{6}\,\frac{\Gamma\left(\frac{1}{6}\right)\Gamma\left(1-\frac{1}{6}\right)}{\Gamma(1)}$$

$$= \frac{1}{6}\,\frac{\pi}{\sin\frac{\pi}{6}} = \frac{1}{6}\cdot\frac{\pi}{\frac{1}{2}} = \frac{\pi}{3}$$

$$\therefore \int_0^1 \frac{dx}{(1-x^6)^{1/6}} = \frac{\pi}{3}.$$

Ex. 8: Prove $\int_0^\infty \sqrt{y}\, e^{-y^2}\,dy \times \int_0^\infty \frac{e^{-y^2}}{\sqrt{y}}\,dy = \frac{\pi}{2\sqrt{2}}$ *2061 Chaitra B. E.*

Solution:

Here, $\int_0^\infty \sqrt{y}\, e^{-y^2}\,dy \times \int_0^\infty \frac{e^{-y^2}}{\sqrt{y}}\,dy$

Put $y^2 = t$, $2y\,dy = dt$, $dy = \frac{dt}{2\,t^{1/2}}$

When $y = 0$, $t = 0$, when $y = \infty$, $t = \infty$

So, $\int_0^\infty \sqrt{y}\, e^{-y^2}\,dy \times \int_0^\infty \frac{e^{-y^2}}{\sqrt{y}}\,dy$

$$= \frac{1}{2}\int_0^\infty \frac{t^{1/4}\,e^{-t}}{t^{1/2}}\,dt \times \frac{1}{2}\int_0^\infty \frac{e^{-t}\,dt}{t^{1/4}\,.\,t^{1/2}}$$

$$= \frac{1}{2}\int_0^\infty t^{\frac{1}{4}-\frac{1}{2}}\,e^{-t}\,dt \times \frac{1}{2}\int_0^\infty e^{-t}\,t^{-\frac{1}{4}-\frac{1}{2}}\,dt$$

$$= \frac{1}{2}\int_0^\infty e^{-t}\,t^{-1/4}\,dt \times \frac{1}{2}\int_0^\infty e^{-t}\,t^{-3/4}\,dt$$

$$= \frac{1}{2}\int_0^\infty e^{-t}\,t^{\frac{3}{4}-1}\,dt \times \frac{1}{2}\int_0^\infty e^{-t}\,t^{\frac{1}{4}-1}\,dt$$

$$= \frac{1}{2}\,\Gamma\left(\frac{3}{4}\right) \times \frac{1}{2}\,\Gamma\left(\frac{1}{4}\right) = \frac{1}{4}\,\Gamma\left(\frac{1}{4}\right)\Gamma\left(1-\frac{1}{4}\right)$$

$$= \frac{1}{4}\,\frac{\pi}{\sin\frac{\pi}{4}} = \frac{\pi}{4.\frac{1}{\sqrt{2}}} = \frac{\pi}{2\sqrt{2}}$$

$$\therefore \int_0^\infty \sqrt{y}\,e^{-y^2}\,dy\,.\int_0^\infty \frac{e^{-y^2}}{\sqrt{y}}\,dy = \frac{\pi}{2\sqrt{2}}\,.$$

Ex. 9: Prove that $\sqrt{\pi}\,\Gamma(2n) = 2^{2n-1}\,\Gamma(n)\,\Gamma\left(n+\frac{1}{2}\right)$

Solution:

We have, $\beta(m, n) = \dfrac{\Gamma(m)\,\Gamma(n)}{\Gamma(m+n)}$ ....(1)

Put m = n, we get

$$\frac{\Gamma(n)\,\Gamma(n)}{\Gamma(2n)} = \beta(n, n) = \int_0^1 (1-x)^{n-1}\,x^{n-1}\,dx$$

Put $x = \sin^2\theta$, $dx = 2\sin\theta\cos\theta\,d\theta$

So,

$$\frac{\Gamma(n)\,\Gamma(n)}{\Gamma(2n)} = \int_0^{\pi/2} (1-\sin^2\theta)^{n-1}\,(\sin^2\theta)^{n-1}\,2\sin\theta\cos\theta\,d\theta$$

$$= 2\int_0^{\pi/2} \cos^{2n-2}\theta\,\sin^{2n-2}\theta\,\sin\theta\cos\theta\,d\theta$$

$$= 2\int_0^{\pi/2} \cos^{2n-1}\theta\,\sin^{2n-1}\theta\,d\theta$$

$$= \frac{2}{2^{2n-1}} \int_0^{\pi/2} (2 \sin\theta \cos\theta)^{2n-1} \, d\theta$$

$$= \frac{2}{2^{2n-1}} \int_0^{\pi/2} (\sin 2\theta)^{2n-1} \, d\theta$$

Put $2\theta = t$, $d\theta = \frac{dt}{2}$

or, $\frac{\Gamma(n)\,\Gamma(n)}{\Gamma(2n)} = \frac{2}{2^{2n-1}} \int_0^{\pi} (\sin t)^{2n-1} \frac{dt}{2} = \frac{1}{2^{2n-1}} \int_0^{\pi} (\sin t)^{2n-1} \, dt$

$\therefore \quad \frac{\Gamma(n)\,\Gamma(n)}{\Gamma(2n)} = \frac{1}{2^{2n-1}} .2 \int_0^{\pi/2} \sin^{2n-1} t \, dt$ ....(2)

Again, put $m = \frac{1}{2}$ in (1), we get

$$\beta\left(\frac{1}{2}, n\right) = \frac{\Gamma\left(\frac{1}{2}\right)\Gamma(n)}{\Gamma\left(n + \frac{1}{2}\right)}$$

or, $\frac{\Gamma\left(\frac{1}{2}\right)\Gamma(n)}{\Gamma\left(n + \frac{1}{2}\right)} = \beta\left(\frac{1}{2}, n\right) = \int_0^1 (1-x)^{\frac{1}{2} - 1} x^{n-1} dx$

$$= \int_0^1 (1-x)^{\frac{-1}{2}} x^{n-1} dx$$

Put $x = \sin^2\theta$, $dx = 2\sin\theta \cos\theta \, d\theta$

So, $\frac{\Gamma\left(\frac{1}{2}\right)\Gamma(n)}{\Gamma\left(n + \frac{1}{2}\right)} = \int_0^{\pi/2} (1 - \sin^2\theta)^{-1/2} (\sin^2\theta)^{n-1} \, 2\sin\theta \cos\theta \, d\theta$

or, $\frac{\sqrt{\pi}\,\Gamma(n)}{\Gamma\left(n + \frac{1}{2}\right)} = 2\int_0^{\pi/2} \sin^{2n-1} \theta \, d\theta$

$\therefore \quad \frac{\sqrt{\pi}\,\Gamma(n)}{2\Gamma\left(n + \frac{1}{2}\right)} = \int_0^{\pi/2} \sin^{2n-1} t \, dt$ ....(3)

From (2) and (3).

$$\frac{\Gamma(n)\,\Gamma(n)}{\Gamma(2n)} = \frac{1}{2^{2n-1}}.2\ \frac{\sqrt{\pi}\,\Gamma(n)}{2\Gamma\left(n+\frac{1}{2}\right)}$$

$$\therefore \quad \sqrt{\pi}\,\Gamma(2n) = 2^{2n-1}\ \Gamma(n)\ \Gamma\left(n+\frac{1}{2}\right)$$

This formula is known as *Duplication Formula.*

Exercise-14

1. Find the value of i. $\Gamma\left(\frac{7}{2}\right)$ ii. $\dfrac{\Gamma\left(\frac{3}{2}\right)\Gamma\left(\frac{5}{2}\right)}{\Gamma(5)}$

2. Evaluate i. $\int_0^{\pi/2} \sin^4 x \cos^2 x\, dx$

   ii. $\int_0^{\pi/4} (1 - 2\sin^2\theta)^{3/2} \cos\theta\, d\theta$

3. Use Beta Gamma function to evaluate

   i. $\int_0^a x^2 (a^2 - x^2)^{3/2}\, dx$ ii. $\int_0^1 x^{3/2} (1 - x)^{3/2}\, dx$

   iii. $\int_0^{2a} x^{9/2} (2a - x)^{-1/2}\, dx$ v. $\int_0^a x^4 \sqrt{a^2 - x^2}\, dx$

   vi. $\int_0^1 \dfrac{x^6\,.\,dx}{\sqrt{1 - x^2}}$ vii. $\int_0^a \dfrac{x^4\, dx}{\sqrt{a^2 - x^2}}$

   viii. $\int_0^a x^3 (a^2 - x^2)^{5/2}\, dx$

5. Using Beta Gamma Function, show that

   i. $\int_0^{\pi/6} \cos^4 3\theta \sin^2 6\theta\, d\theta = \dfrac{5\pi}{192}$ 2070 *Ashad B. E.*

   ii. $\int_0^{\pi/6} \cos^2 6\theta \sin^4 3\theta\, d\theta = \dfrac{7\pi}{192}$

   iii. $\int_0^{\pi} \sin^6 \dfrac{x}{2} \cos^8 \dfrac{x}{2} = \dfrac{5\pi}{2^{11}}$ 2069 *Ashad B. E.*

iv. $\int_0^{\pi/4} \sin^4 x \cos^2 x \, dx = \frac{3\pi - 4}{192}$

6. Prove that $\int_0^{\infty} x^2 e^{-x^4} dx \times \int_0^{\infty} e^{-x^4} dx = \frac{\pi}{8\sqrt{2}}$ 2062 Baishakh B. E.

7. Prove that $\Gamma\left(\frac{1}{3}\right) \Gamma\left(\frac{2}{3}\right) = \frac{2\pi}{\sqrt{3}}$

8. Show that $\beta(m, n)\, \beta(m + n, l) = \beta(n, l)\, \beta(n + l, m)$

9. Show that $\int_0^{\infty} e^{-x^2} x^{\alpha} dx = \frac{1}{2} \Gamma\left(\frac{\alpha + 1}{2}\right)$

10. Prove that

$$\int_a^b (x - a)^m (b - x)^n dx = (b - a)^{m+n+1} \frac{\Gamma(m + 1)\, \Gamma(n + 1)}{\Gamma(m + n + 2)}, m > -1, n > -1$$

[Hint: Put $x - a = (b - a)\, t$]

Answers

1. i. $\frac{15}{8}\sqrt{\pi}$ ii. $\frac{\pi}{64}$ 2. i. $\frac{\pi}{32}$ ii. $\frac{3\pi}{16\sqrt{2}}$

3. i. $\frac{\pi a^6}{32}$ ii. $\frac{3\pi}{128}$ iii. $\frac{63\pi a^5}{8}$ iv. $\frac{\pi a^6}{32}$

v. $\frac{5\pi}{32}$ vi. $\frac{3\pi a^4}{16}$ vii. $\frac{2a^9}{63}$